

[illegible]

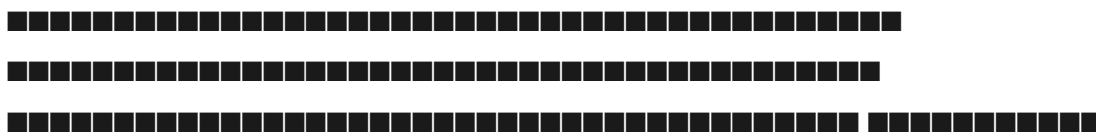
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The first observation: in agricultural field trials, results obtained from experimental plots fail to apply to actual farmland far more often than they succeed. This is not a failure of experimental precision — it is a structural consequence of the fact that each system carries its own accumulated load history and its

own residual capacity. Averages cannot substitute for instance-specific state.



The second observation: structural systems — whether a blood vessel, a dam, or a bridge — do not fail suddenly from a state of apparent safety. They approach failure gradually through accumulation, while appearing superficially normal. The rate of change may be near zero at the moment of highest risk. What matters is not how fast the system is changing, but how close it currently stands to its structural limit.



From these two observations, we introduce the diagnostic ratio:



$$R = \delta / \tau$$

R takes values from 0 to 1.0. R = 1.0 is the fracture point — the moment of structural transition, collapse, or biological failure. The safety boundary value is placed at R = 0.85: beyond this point, the system may reach R = 1.0 through a single additional stress event. The interval from R = 0.85 to R = 1.0 does not represent remaining margin. It represents an acceleration zone in which intervention may no longer be sufficient.



This ratio is not a predictor of failure time. It is the minimal diagnostic form for expressing how close an instance-specific system currently stands to exhaustion of its residual capacity.

The threshold-based computational approach presented here is broadly applicable across physical, biological, engineered, and computational systems. The theoretical foundations are documented in the NRA-IDE Project [DOI: 10.5281/zenodo.19420854].



1.1 The Problem of Averages in Safety Management

It is widely recognized among agricultural practitioners that the results of field trials conducted on experimental plots fail to apply to actual farmland far more often than they succeed. A trial conducted on three plots of 10 ares each does not reliably predict what will happen on a one-hectare field.

[illegible]

This is not a failure of experimental rigor. The trial is conducted correctly. The measurements are accurate. The problem lies elsewhere: each plot and each actual field carries its own accumulated load history and its own residual capacity. Their instance-specific states differ. The average of the trial plots is a statistical abstraction that may correspond to none of the actual fields.

The same structural fact applies beyond agriculture. A patient is not the average patient. A dam is not the average dam of its design class. A bridge is not the average bridge of its construction year. Each carries its own accumulated deviation and its own residual capacity. Safety management that places averages at the center leaves the overwhelming majority of systems — those that differ from the average in structurally significant ways — outside the boundary of effective diagnosis.

[illegible]

1.2 Accumulation Is Not Visible in Rate-of-Change Monitoring

1.2 ■■■■■■■■■■

Consider a blood vessel. Cholesterol deposits, calcification, and chemical degradation accumulate over years. On any given day, the rate of change may be near zero. Blood pressure readings may be within normal range. Surface indicators show nothing unusual. Yet the proximity to rupture — the ratio of accumulated structural load to remaining vessel capacity — may be approaching a critical value.

[illegible]

Rate-of-change monitoring answers the question: "how fast is this changing?" Proximity monitoring answers the question: "how close to the limit is this right now?" These are different questions. They require different measurements. And they produce different safety outcomes in systems where the most dangerous state is one of apparent stillness.

[illegible]

1.3 A Different Methodological Basis within Safety Engineering

1.3

Existing safety engineering encompasses both post-failure analysis — the investigation of cause after collapse — and pre-failure monitoring — the observation of structural state before collapse. Both have value. Both have their place within the broader discipline.

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The distinction this paper draws is methodological. The diagnostic framework proposed here — $R = \delta/\tau$ — is based on a different set of indicators than conventional rate-of-change or threshold-exceedance monitoring. Where conventional monitoring uses the instantaneous rate and the distance from a

statistical mean, this framework uses the accumulated deviation and the ratio to remaining capacity. The difference in what is measured leads to a difference in what can be diagnosed.

Specifically, $R = \delta/\tau$ makes visible what rate-based monitoring cannot see: the approach to structural limit during periods of apparent stability. This property is not a replacement for existing methods. It is a different diagnostic layer that answers a question existing methods do not ask.

R = δ/τ

1.4 Scope of This Paper

1.4 ■■■■■

This paper introduces $R = \delta/\tau$ as a diagnostic ratio, defines its three formulations (fundamental, asymmetric double-fluctuation, and hybrid complement), specifies the design principles that govern its use (Fail-Closed, Disclosure Axiom, Instance-Specific Principle), and demonstrates its structural consistency across multiple domains through documented case studies.

R = δ/τ

Fail-Closed

2. Distinction from Existing Approaches / ■■■■■■■■■■

2.1 ■■■■■■■■■■ / Rate vs. Proximity

[illegible]

R 0 1.0

R		
R < 0.4		
0.4 0.75		
0.75 0.85		
0.85 1.0		
R = 1.0		

3.2 / Safety Boundary Value

R = 1.0 R = 1.0

R = 0.85 R = 1.0 0.85 1.0

R = 0.85 R = 1.0

R = 0.85 Safety Boundary Value

Fail-Closed =

R = 1.0 R = 0.85

R = 1.0

0.7

R = 1.0

R = 1.0 R = 1.0

[illegible]

A system that has crossed $R = 1.0$ cannot return to its prior state. This irreversibility is what distinguishes $R = 1.0$ from a simple warning threshold. It is not "the limit from which return is possible." It is "the boundary beyond which return is not."

This irreversibility is what makes the safety boundary value ($R = 0.85$) more critical than $R = 1.0$ itself. Recognition at $R = 1.0$ is already too late.

Dynamic τ : Asymmetric Double-Fluctuation Formula and Survival Zone

■■■■■ - ■■■■■■■■■■τ■■: 37.5°C■■ → ■■■■■■ - ■■■■■■■■■■τ■■: 35°C■■ → ■■■■ -
■■■■■■■■■■■■■■

■■■■■■■■■■■2■■■■■■

[illegible]

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3.4 Floating-Point Error and Precision

1. $\delta \tau$

$0.1 + 0.2 = 0.30000000000000004$

- 1: ϵ
- 2: ϵ
- 3: $\epsilon + \epsilon$
- n :
- $\rightarrow$

AI

Goldberg, 1991

IDE

Intensional Dynamics Engine IDE

IDE

- 1: $\epsilon \rightarrow$
- 2: $\epsilon \rightarrow \epsilon \rightarrow$
- 3: $\epsilon \rightarrow \epsilon \rightarrow$
- n :
- $\rightarrow$

1

IDE

Integer Phase Lock

3.5 IDE / Hybrid Complement Formula

$$\frac{d^2x}{dt^2} + \gamma \dot{x} = F_{\text{IDE}}(x) + G(r) \cdot \Phi(x)$$

$$G(r) = r \cdot \frac{1}{k + |r|}, \quad \text{quad } r = x_{\text{IDE}} - x$$

- γ
- F_{IDE}
- $G(r)$

$G(r)$

$$|r| \ll k \quad G(r) \approx r^2/k \text{ --- IDE}$$

$$|r| \gg k \quad G(r) \approx r \text{ ---}$$

$$R = 0.85 \quad R = 0.87$$

$G(r)$

- $\rightarrow$ 1: $R = \delta/\tau$
- $\rightarrow$ 2:
- $\rightarrow \text{IDE}$

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4. Design Principles /

4.1 Fail-Closed

δ	
τ	
R = 0.85	
R = 1.0	

Correlated Threshold Effect

δ R = 0.85

A $\tau = 140$ B $\tau = 60$ 30
 50 R 50 B

5.2 / Dam

δ	
τ	
	R = 0.85 1.0
R = 1.0	

1963 1,917
R 2,000

5.3 / Structures

2013 1,134

- 2003  14% 24%
- 2008  15.7m 24%
- 2009  27% 7%

The instruments exist. The monitoring infrastructure exists.

What has been absent is the framework connecting measured δ and τ to the Safety Boundary Value R as a Fail-Closed activation condition. This paper proposes that connecting structure.

The Safety Boundary Value is not a fixed regulatory figure. As measurement data accumulates, the threshold is refined as an instance-specific value. Continuous observation updates the threshold. This is the practical form of instance-specific safety management.

7. Conclusion / ■■

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R = δ/τ R = 1.0R = 0.85R
 $= 0.85$

Fail-Closed

XX

R

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References / ■■■■

Goldberg, D. (1991). What every computer scientist should know about floating-point arithmetic. *ACM Computing Surveys*, 23(1), 5–48.

Lenton, T.M., Held, H., Kriegler, E., Hall, J.W., Lucht, W., Rahmstorf, S., & Schellnhuber, H.J. (2008). Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences*, 105(6), 1786–1793.

M-Tokuni (2026). *NRA-IDE v1.0: Nomological Ring Axioms — Intensional Dynamics Engine*. Zenodo. <https://doi.org/10.5281/zenodo.19420854>

National Diet of Japan (2012). *The Official Report of the Fukushima Nuclear Accident Independent Investigation Commission*. The National Diet of Japan.

Paris, P. & Erdogan, F. (1963). A critical analysis of crack propagation laws. *Journal of Basic Engineering*, 85(4), 528–533.

Perera, D., Smakhtin, V., Williams, S., North, T., & Curry, A. (2021). *Ageing Water Infrastructure: An Emerging Global Risk*. UNU-INWEH.

Scheffer, M., Bascompte, J., Brock, W.A., Brovkin, V., Carpenter, S.R., Dakos, V., Held, H., van Nes, E.H., Rietkerk, M., & Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature*, 461, 53–59.

Shirzaei, M. et al. (2025). Aging dams, political instability, poor human decisions and climate change: recipe for human disaster. *npj Natural Hazards*. <https://doi.org/10.1038/s44304-024-00056-1>

Government Investigation Commission on the Fukushima Nuclear Power Station Accident (2012). *Final Report*. Tokyo.

International Atomic Energy Agency (2015). *The Fukushima Daiichi Accident: Report by the Director General*. IAEA.

Sohn, H., Farrar, C.R., Hemez, F.M., Shunk, D.D., Stinemates, D.W., Nadler, B.R., & Czarnecki, J.J. (2004). *A Review of Structural Health Monitoring Literature: 1996–2001*. Los Alamos National Laboratory.

Acknowledgments / ■■

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